E16: Linear Spectral Concentration and Convergence of the Dominant Fourier Wavenumber

Manuscript 1 — Self-Emergent Fourier Cymatics
A self-contained proof with explicit hypotheses and counterexample

October 26, 2026

Abstract

For the linearized finite-band evolution $\partial_t A = (r - a\Delta - b\Delta^2)A$ on $\mathbb{R}^d$, with $a, b > 0$, we establish the exact spectral-power law $P(\xi, t) = P(\xi, 0)e^{2\lambda(\xi)t}$, where $\lambda(\xi) = r + a\,|\xi|^2 - b\,|\xi|^4$. Its unique maximizing *radius* is $k_\star = \sqrt{a/(2b)}$. If the initial Fourier power has positive mass in every neighborhood of $\{|\xi| = k_\star\}$, the normalized spectral measure concentrates exponentially in any fixed tubular neighborhood of that shell. Under an additional continuous, vanishing-at-infinity hypothesis on the initial spectral power, every maximizing Fourier wavenumber tends radially to $k_\star$. We also give a quantitative $O(t^{-1/2})$ bound when a shell frequency is initially populated, a radial-spectrum formulation, a torus qualification, and a counterexample showing why the initial-support hypothesis is necessary. The result is linear spectral selection, not a nonlinear localization or confinement theorem.

1 Model, notation, and hypotheses

Fix a spatial dimension $d \geq 1$, $a, b > 0$, and $r \in \mathbb{R}$. Consider

$$\partial_t A = (r - a\Delta - b\Delta^2)A, \qquad A(0) = A_0 \in L^2(\mathbb{R}^d; \mathbb{C}), \quad A_0 \neq 0. \tag{1}$$

The unitary Fourier transform is

$$\widehat{A}(\xi) = (2\pi)^{-d/2} \int_{\mathbb{R}^d} e^{-ix\cdot\xi} A(x)\,\mathrm{d}x, \tag{2}$$

initially on Schwartz functions and then extended by Plancherel. Put

$$k_\star = \sqrt{\frac{a}{2b}} > 0, \qquad \mu = r + \frac{a^2}{4b}, \tag{3}$$

$$\lambda(\xi) = r + a\,|\xi|^2 - b\,|\xi|^4 = \mu - b\big(|\xi|^2 - k_\star^2\big)^2, \qquad q(\xi) = \big(|\xi|^2 - k_\star^2\big)^2. \tag{4}$$

The preferred shell is

$$\mathcal{S}_\star = \{\xi \in \mathbb{R}^d : |\xi| = k_\star\}.$$

The relevant spectral-support hypothesis is

$$\int_{\{\xi:\,||\xi| - k_\star| < \varepsilon\}} \left|\widehat{A}_0(\xi)\right|^2 \mathrm{d}\xi > 0 \qquad \text{for every } \varepsilon > 0. \tag{H}$$

This is an *essential-support* condition: it does not require nonzero pointwise power on the measure-zero sphere itself.

2 Exact Fourier evolution

Proposition 2.1 (Exact modal power law)**.** *For every $A_0 \in L^2(\mathbb{R}^d)$, equation (1) has a unique mild solution $A \in C([0,\infty); L^2(\mathbb{R}^d))$, and*

$$\widehat{A}(t,\xi) = e^{\lambda(\xi)t}\widehat{A}_0(\xi), \tag{5}$$

$$P(\xi,t) := \left|\widehat{A}(t,\xi)\right|^2 = P(\xi,0)e^{2\lambda(\xi)t} \quad \text{for a.e. } \xi \in \mathbb{R}^d. \tag{6}$$

Proof. Using $\widehat{\Delta A} = -|\xi|^2 \widehat{A}$ and $\widehat{\Delta^2 A} = |\xi|^4 \widehat{A}$, Fourier transformation gives, frequency by frequency, $\partial_t \widehat{A} = \lambda(\xi)\widehat{A}$. Solving this scalar ODE yields (5). Since $\lambda(\xi) \le \mu$, $\left|e^{t\lambda(\xi)}\right| \le e^{t\mu}$ for $t \ge 0$, so multiplication by $e^{t\lambda}$ is bounded on L^2. The semigroup law is immediate, and dominated convergence yields strong continuity at $t = 0$ (and hence at all $t \ge 0$). The Fourier multiplier therefore defines the unique L^2 mild solution generated by $r - a\Delta - b\Delta^2$. Taking absolute values squared proves (6). □

Lemma 2.2 (Unique maximizing radius and strict gap)**.** *The set of Fourier vectors maximizing λ is exactly $\mathcal{S}_\star$. For every $\delta > 0$,*

$$\big||\xi| - k_\star\big| \ge \delta \quad \Longrightarrow \quad \mu - \lambda(\xi) \ge bk_\star^2\delta^2. \tag{7}$$

Proof. Completion of the square in (4) shows $\lambda \le \mu$, with equality exactly when $|\xi| = k_\star$. Furthermore,

$$\mu - \lambda(\xi) = b(|\xi| - k_\star)^2(|\xi| + k_\star)^2 \ge bk_\star^2(|\xi| - k_\star)^2.$$

This proves (7). □

3 Concentration of normalized spectral power

The integral $Z(t) := \int_{\mathbb{R}^d} P(\xi,t)\,\mathrm{d}\xi$ is finite and strictly positive for every finite $t \ge 0$. Define the probability measure

$$\nu_t(E) = \frac{\int_E P(\xi,t)\,\mathrm{d}\xi}{Z(t)} \quad \text{for measurable } E \subset \mathbb{R}^d. \tag{8}$$

Theorem 3.1 (E16: concentration on the preferred shell)**.** *Suppose $A_0 \neq 0$ belongs to $L^2(\mathbb{R}^d)$ and satisfies (H). For every $\delta > 0$, let*

$$E_\delta = \{\xi : \big||\xi| - k_\star\big| \ge \delta\}.$$

Then there exists $C_\delta \in (0,\infty)$, depending only on the initial spectrum and δ, such that

$$\boxed{\nu_t(E_\delta) \le C_\delta e^{-bk_\star^2\delta^2 t} \xrightarrow[t\to\infty]{} 0.} \tag{9}$$

Equivalently, for every $\delta > 0$,

$$\lim_{t\to\infty} \nu_t\big(\{\xi : \big||\xi| - k_\star\big| < \delta\}\big) = 1.$$

Proof. Write $f(\xi) = \left|\widehat{A}_0(\xi)\right|^2 \in L^1(\mathbb{R}^d)$, $F = \int_{\mathbb{R}^d} f \,\mathrm{d}\xi = \|A_0\|_{L^2}^2 > 0$, and $m_\delta = k_\star^2\delta^2 > 0$. By (4) and (6),

$$P(\xi, t) = e^{2\mu t} f(\xi) e^{-2btq(\xi)}. \tag{10}$$

On E_δ we have $q \geq m_\delta$ by Lemma 2.2, so

$$\int_{E_\delta} P(\xi, t) \,\mathrm{d}\xi \leq e^{2\mu t} e^{-2bm_\delta t} F. \tag{11}$$

Consider the open inner region

$$B_\delta = \{\xi : q(\xi) < m_\delta/2\}.$$

Because q is continuous and vanishes on $\mathcal{S}_\star$, B_δ contains some tubular neighborhood $\{||\xi| - k_\star| < \varepsilon_\delta\}$. Hypothesis (H) therefore implies

$$c_\delta := \int_{B_\delta} f(\xi) \,\mathrm{d}\xi > 0.$$

For $\xi \in B_\delta$, $e^{-2btq(\xi)} \geq e^{-bm_\delta t}$, whence

$$Z(t) \geq e^{2\mu t} e^{-bm_\delta t} c_\delta. \tag{12}$$

Dividing (11) by (12),

$$\nu_t(E_\delta) \leq \frac{F}{c_\delta} e^{-bm_\delta t}.$$

Taking $C_\delta = F/c_\delta$ and $m_\delta = k_\star^2\delta^2$ proves the result. □

Remark 3.2 (Normalized does not mean persistent amplitude). The theorem concerns P/Z, not the magnitude of $Z(t)$. In particular, if $\mu = 0$, then $P(\xi, t) \to 0$ for almost every ξ, since the exact shell has Lebesgue measure zero; dominated convergence gives $\|A(t)\|_2 \to 0$. Nevertheless ν_t still concentrates on the shell under (H).

4 Convergence of the spectral argmax

For L^2 data alone, the Fourier transform and its pointwise maximum are not intrinsically defined. The next theorem states the additional regularity needed for a pointwise argmax.

Theorem 4.1 (Argmax convergence)**.** *In addition to the hypotheses of Theorem 3.1, assume that the initial power $f(\xi) = \left|\widehat{A}_0(\xi)\right|^2$ has a nonnegative continuous representative in $C_0(\mathbb{R}^d)$ (continuous and vanishing at infinity). For every $t \geq 0$, the set $\arg\max_{\xi\in\mathbb{R}^d} P(\xi, t)$ is nonempty. Every choice $\xi_t \in \arg\max P(\cdot, t)$ satisfies*

$$\boxed{\lim_{t\to\infty} |\xi_t| = k_\star.} \tag{13}$$

This is radial convergence and does not select one vector direction.

Proof. Since $f \in C_0$, $f \geq 0$, and $f \not\equiv 0$, $M := \sup_\xi f(\xi)$ lies in $(0, \infty)$. For each finite t, (10) defines a nonzero nonnegative function in $C_0(\mathbb{R}^d)$; hence its positive global maximum is attained.

Fix $\delta > 0$ and set $m_\delta = k_\star^2\delta^2$. Since (H) holds and q vanishes continuously on the preferred shell, there exists $\eta_\delta \in \mathbb{R}^d$ such that

$$f(\eta_\delta) > 0, \qquad q(\eta_\delta) < m_\delta/2. \tag{14}$$

At this point,

$$P(\eta_\delta, t) \geq e^{2\mu t} f(\eta_\delta) e^{-bm_\delta t}. \tag{15}$$

For every $\xi \in E_\delta$,

$$P(\xi, t) \le e^{2\mu t} M e^{-2bm_\delta t}. \tag{16}$$

Combining the bounds yields

$$\frac{\sup_{\xi \in E_\delta} P(\xi, t)}{P(\eta_\delta, t)} \le \frac{M}{f(\eta_\delta)} e^{-bm_\delta t} \xrightarrow[t\to\infty]{} 0. \tag{17}$$

Thus for all sufficiently large t the power at η_δ exceeds the power at every $\xi \in E_\delta$. Every global maximizer ξ_t then lies outside E_δ, i.e., $||\xi_t| - k_\star| < \delta$. As $\delta > 0$ was arbitrary, (13) follows. □

Corollary 4.2 (Explicit $t^{-1/2}$ upper bound)**.** *Under the hypotheses of Theorem 4.1, suppose there is $\eta_\star \in \mathcal{S}_\star$ with $f(\eta_\star) > 0$. Let $M = \sup_\xi f(\xi)$. Then, for every $t > 0$ and every $\xi_t \in \arg\max P(\cdot, t)$,*

$$\boxed{||\xi_t| - k_\star| \le \sqrt{\frac{\log(M/f(\eta_\star))}{2bk_\star^2 t}}.} \tag{18}$$

In particular the radial argmax error is $O(t^{-1/2})$.

Proof. Because $q(\eta_\star) = 0$ and ξ_t maximizes P,

$$f(\xi_t) e^{-2btq(\xi_t)} \ge f(\eta_\star).$$

Since $f(\xi_t) \le M$,

$$q(\xi_t) \le \frac{\log(M/f(\eta_\star))}{2bt}.$$

The factorization $q(\xi_t) = (|\xi_t| - k_\star)^2(|\xi_t| + k_\star)^2$ and $|\xi_t| + k_\star \ge k_\star$ give (18). □

5 Radial spectral density: the scalar wavenumber formulation

In dimension $d > 1$, a Fourier frequency is a vector ξ; k usually denotes its nonnegative radius. Define, for a.e. $k > 0$,

$$\mathscr{P}(k, t) = k^{d-1} \int_{\mathbb{S}^{d-1}} P(k\omega, t)\, \mathrm{d}\omega. \tag{19}$$

For $d = 1$, the integral is over the two-point sphere $\mathbb{S}^0 = \{-1, +1\}$ with counting measure. Polar integration gives $\int_0^\infty \mathscr{P}(k, t)\, \mathrm{d}k = Z(t)$. Because $\lambda(k\omega) = \mu - b(k^2 - k_\star^2)^2$,

$$\boxed{\mathscr{P}(k, t) = \mathscr{P}(k, 0) e^{2[\mu - b(k^2 - k_\star^2)^2]t}} \quad \text{for a.e. } k > 0. \tag{20}$$

Corollary 5.1 (Radial concentration and maximizing wavenumber)**.** *Assume (H). Then for every $\delta > 0$,*

$$\frac{\int_{\{k \ge 0 : |k - k_\star| \ge \delta\}} \mathscr{P}(k, t)\, \mathrm{d}k}{\int_0^\infty \mathscr{P}(k, t)\, \mathrm{d}k} \longrightarrow 0. \tag{21}$$

If, in addition, $\mathscr{P}(\cdot, 0)$ has a nonnegative, nonzero representative in $C_0([0, \infty))$, then every choice

$$k_t \in \arg\max_{k \ge 0} \mathscr{P}(k, t)$$

satisfies $k_t \to k_\star$. If $\mathscr{P}(k_\star, 0) > 0$, the estimate (18) holds with $f(\eta_\star)$ and M replaced by $\mathscr{P}(k_\star, 0)$ and $\sup_{k \ge 0} \mathscr{P}(k, 0)$, respectively.

Proof. Equation (21) is precisely Theorem 3.1 in polar coordinates. For the argmax, apply the comparison argument in the proof of Theorem 4.1 to the scalar function $\mathscr{P}(k, t)$, using (20). The quantitative bound follows by comparing the maximum to the positive value at $k_\star$, exactly as in Corollary 4.2. □

6 Counterexample and periodic-domain qualification

Proposition 6.1 (Failure without spectral access)**.** *There exist nonzero* $A_0 \in \mathcal{S}(\mathbb{R}^d)$ *for which* $\arg\max P(\cdot, t)$ *never approaches* $\mathcal{S}_\star$*.*

Proof. Choose $\delta > 0$ and nonzero $\widehat{A}_0 \in C_c^\infty(\mathbb{R}^d)$ supported in $\{\xi : |\xi| > k_\star + \delta\}$. Its inverse Fourier transform is Schwartz. Since the exact multiplier formula (5) preserves Fourier support, $P(\xi, t)$ vanishes whenever $|\xi| \le k_\star + \delta$ for every $t \ge 0$. Each $P(\cdot, t)$ has a positive global maximum and all its maximizers satisfy $|\xi_t| > k_\star + \delta$. Thus $|\xi_t|$ cannot converge to $k_\star$. □

Remark 6.2 (Flat tori). On the torus $\mathbb{T}_L^d$ with Fourier lattice $\xi_n = (2\pi/L)n$, $n \in \mathbb{Z}^d$, the exact coefficientwise law is

$$P_n(t) = P_n(0)e^{2\lambda(\xi_n)t}.$$

The dominant asymptotic modes belong to the set of initially populated lattice frequencies maximizing $\lambda(\xi_n)$. Because the lattice is discrete, this need not contain a point with $|\xi_n| = k_\star$. Therefore convergence of the selected lattice radius to the continuum $k_\star$ is not automatic for a fixed torus.

7 Conclusion and scope

The precise E16 result is the exact modal-power identity

$$\boxed{P(\xi, t) = P(\xi, 0)e^{2\lambda(\xi)t}}$$

and, under essential spectral access to the maximizing shell, normalized concentration near $|\xi| = k_\star$. Convergence of the pointwise spectral argmax requires an additional condition ensuring that a maximum exists and controlling the initial spectral amplitude; C_0 regularity is sufficient. For a scalar radial spectrum, the corresponding statement is $\arg\max_{k \ge 0} \mathscr{P}(k, t) \to k_\star$ under the hypotheses of Corollary 5.1.

The theorem does not imply a nonzero whole-space bound state, nonlinear pattern persistence, phase or directional selection, or spatial confinement. No such conclusion follows solely from the linear Fourier multiplier.

Canonical model source. The equations correspond to E12–E16 in WCT Full Equation List (Corrected).